Poster #433

MetaboFood-KDB: A Cloud Knowledgebase for Searching Metabolomics and Exposomics Data for Nutritionally Relevant Compounds

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Introduction

Untargeted metabolomics and exposomics based on liquid chromatography-mass spectrometry (LC-MS) is being increasingly utilized in nutrition intervention and biomarker studies. The resulting LC-MS(/MS) signals need to be assigned to food compounds and their metabolites for researchers to interpret data. This assignment relies on the availability of databases of nutritionally relevant compounds and their metabolites. Such studies will also benefit greatly from an informatics platform that can automatically search LC-MS(/MS) signals against such databases that allow researchers to interactively explore nutritionally relevant compounds in various foods and their metabolites.

To address this need, we are building a cloud knowledgebase named MetaboFood-KDB. Currently, it features a library of (P-MetDB) of phytochemicals and their metabolites derived from comprehensive literature reviews and careful curation of compounds from 17 food. In building P-MetDB, we aim to overcome the reported issue of inaccuracies and errors in existing food databases. For many of the compounds, LC- MS/MS spectra of the compounds and their adduct forms are provided. All of these spectra are acquired from pure reference standards on a UPLC-Orbitrap platform. Other nutritionally relevant compounds and their metabolites will gradually be added.

To search LC-MS(/MS) signals against MetaboFood-KDB, users will upload a table of features/peaks and/or LC-MS/MS spectra that have been exported by software tools that process raw LC-MS(/MS) data. After completing the database search, users can take advantage of the built-in visual analytics capabilities to visually and interactively explore MetaboFood-KDB.

P-MetDB and Beyond

Details about P-MetDB can be found on Poster #92.

- A phytochemical metabolome database built to support metabolomics and exposomics research.
- Every compound in it has gone through a systematic review.
- Currently covers coffee, tea, grapefruit, lemon, organce, apple, grains, banana, berries, cocoa, and onion.
- Chemical composition and metabolite identities extracted food databases, Pubchem, and Web of Science.
- LC-MS/MS spectra have been obtained from pure reference standards.
- MetaboFood-KDB will be constantly expanding to cover more and more compound classes and foods.

Cloud Infrastructure of MetaboFood-KDB

- Similar to the cloud infrastructure of ADAP-KDB at https://www.adap.cloud that was designed for searching, tracking, and prioritizing LC-MS and GC-MS spectra.
- Compound identification and annotation will be based first on monoisotopic mass.
- Retention time and LC-MS/MS spectra will be taken into consideration whenever available.

Figure 1: Display of results after matching query/experimental LC-MS/MS against spectra in the spectral library of physical reference standards.

Generation of LC-MS/MS Reference Spectra

The phytochemicals that were selected from systemic literature review and verified on LC-MS/MS qualitative analysis were dissolved in methanol/water/DMSO depending on solubility. Standard mixtures with a final concentration at 500 ng/ml for each compound were prepared by mixing 7-12 individual stock solutions with water-methanol (95:5). Isomeric compounds were prepared separately in individual solutions. A 5 µL aliquot of each standard mixture was injected for the UPLC-HRMS analysis. Chromatographic and HRMS data were acquired on a Vanquish UHPLC system coupled to a Q ExactiveTM HF-X Hybrid Quadrupole-Orbitrap Mass Spectrometer (Thermo Fisher Scientific, San Jose, CA). The chromatographic data were acquired via an HSS T3 C18 column (2.1 x 100 mm, 1.7 µm, Waters Corporation) at 50 °C with binary mobile phases of water (A) and methanol (B), each containing 0.1% formic acid (v/v). The linear gradient consisted of an initial composition of 2% B, increased to 100% B over 16 min, and was held at 100% B for 4 min, with a flow rate at 0.4 ml/min. The spectral data was acquired from 70 to 1050 m/z using data-dependent scanning mode in positive and negative mode. Progenesis QI (version 2.1, Waters Corporation) was used for peak picking, data extraction (RT, MS, and MS/MS) for files used for library construction. Compound information, including structure, formula, monoisotopic exact mass, chemical ID, et al., were extracted from SDF files directly downloaded from Pubchem, and/or HMD.

Visual Analytics

- Overall similarities among different foods and uniqueness of each food in terms of chemical compositions are displayed using a Self-Organizing Map (SOM) (Figure 2).
- More detailed comparison of different foods is displayed using a Sankey diagram (Figure 3) that can be interactively explore by selecting various criterion (e.g, for certain compound classes).
- The visual analytics capabilities can also be used for diet design.

Figure 2: SOM.

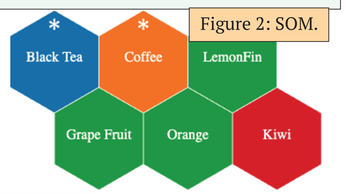
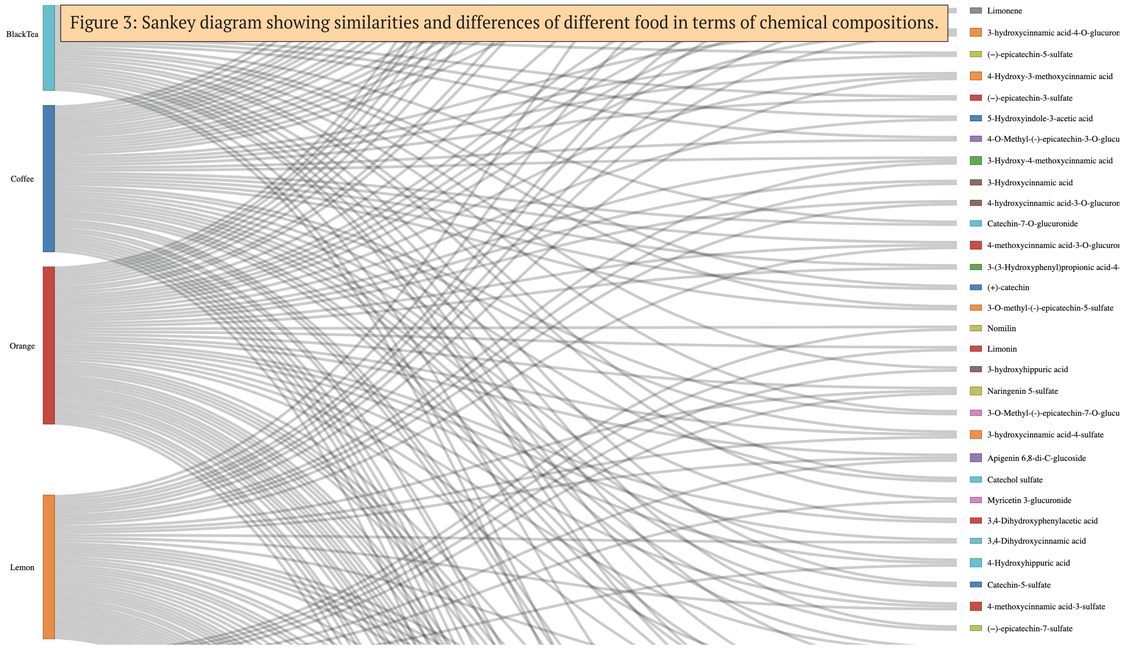


Figure 3: Sankey diagram showing similarities and differences of different food in terms of chemical compositions.



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References

Aleksandr Smirnov, Yunfei Liao, Eoin Fahy, Shankar, Subramaniam, Xiuxia Du: ADAP-KDB: **A Spectral Knowledgebase for Tracking and Prioritizing Unknown Mass Spectra in the NIH's Metabolomics Data Repository.** Manuscript in revision.